An Introduction to Z-Spin Cosmology

A Zero-Free-Parameter Cosmology from Pure Geometry

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This document introduces the Z-Spin Cosmology research programme (55 papers).
Every numerical claim is accompanied by observational cross-checks and pre-registered falsification conditions.

Verification: 55 Papers | ~155 Gates, ~1,466+ Tests PASS | Zero Free Parameters

§0. Abstract

Z-Spin Cosmology is a new ontological and mathematical framework that unifies macroscopic gravity (ΛCDM) and microscopic quantum mechanics (the Standard Model) without introducing any phenomenological free parameters.

The universe is proposed to possess a three-layered architecture: a macroscopic space/particle sector (**X**, dim = 3), a microscopic time/wave sector (**Y**, dim = 6), and a Planck-scale entanglement boundary that mediates between them (**Z**, dim = 2). From the polyhedral geometric asymmetries of these sectors, a universal 'geometric impedance' **A = 35/437** is derived. This single geometric ratio yields predictions—inflation, the Hubble tension, matter density, baryon asymmetry, gauge coupling constants, and the Born rule—spanning sixty orders of magnitude, all with zero adjustable parameters.

Epistemic Status Legend

Status	Definition
PROVEN	Established by mathematical proof from axioms and definitions alone.
DERIVED	Deduced from A = 35/437 and Q = 11 with zero free parameters.
DERIVED-CONDITIONAL	Derived under stated conditions (e.g. 1-loop, asymptotic safety).
VERIFIED	Numerically confirmed (50-digit precision).
OBSERVATION	Consistent with observational data; theoretical derivation pending.
TESTABLE / HYPOTHESIS	Awaiting future experimental test / supported by strong circumstantial evidence.
NON-CLAIM / RETRACTED	Explicitly outside scope / Withdrawn (documented as scientific progress).

§1. Overture: Why Geometry?

Modern physics rests upon two pillars—quantum field theory and general relativity—each spectacularly successful within its own domain. Yet at their intersection, six fundamental crises persist: (1) the Hubble tension—a 5σ discrepancy between CMB and local measurements of the expansion rate; (2) the S_8

tension—disagreement in the growth rate of cosmic structure; (3) the hierarchy problem—why the Higgs mass is 10^{32} times lighter than the Planck scale; (4) dark matter—85% of the universe's mass budget remains unidentified; (5) dark energy—the unknown agent accelerating cosmic expansion; and (6) the measurement problem—the origin of the Born rule and wavefunction collapse in quantum mechanics.

Z-Spin Cosmology proposes that these six crises are not independent puzzles but symptoms of a single geometric origin. Rather than introducing new particles or new free parameters, the framework derives existing physics from the *shape* of spacetime.

§2. The Geometric Stage: From One Constant to Everything

2.1. The Three-Sector Universe and the Q = 11 Architecture

The universe decomposes into three sectors. The **X**-sector (dim = 3) encodes macroscopic space and particle degrees of freedom. The **Y**-sector (dim = 6) encodes microscopic time and wave dynamics. The **Z**-sector (dim = 2) is the Planck-scale entanglement boundary that mediates between them. The total dimensionality is **Q = 2 + 3 + 6 = 11**, coinciding exactly with the connector space of j = 1/2 quantum tetrahedra in loop quantum gravity (Theorem 5.1, PROVEN). The gauge dimension is $G = 12 = \dim(su(3)\oplus su(2)\oplus u(1)) = Q + 1$ (PROVEN).

2.2. Derivation of the Geometric Impedance A = 35/437

The X-sector mediator—the truncated octahedron (V = 24, F = 14)—and the Y-sector mediator—the truncated icosahedron (V = 60, F = 32)—each carry a duality-deviation invariant:

$$\delta_X = |V-F|/(V+F) = 10/38 = 5/19$$

$$\delta_Y = |V-F|/(V+F) = 28/92 = 7/23$$

$$\mathbf{A = \delta_X \times \delta_Y = (5/19) \times (7/23) = 35/437 \approx 0.0800915}$$

The δ-uniqueness theorem (ZS-F2, Theorem 3.1, PROVEN) establishes that $\delta(P) = |V-F|/(V+F)$ is the *unique* function satisfying seven physically necessary axioms. Three independent convergence paths—the Regge 1-loop calculation, the asymptotic-safety NGFP coupling $\xi^* \approx 0.08$, and the β-function spectral identity—all yield the same value. ZS-F6 v1.0 proves that this convergence is necessary: the unique 1-loop self-consistency fixed point of the Regge lattice heat kernel is ξ_quantum = δ_X·δ_Y = 35/437 = A [DERIVED, 20/20 PASS].

2.3. The Core Action: A Single Equation

$$\mathbf{S = \int d^4x \sqrt{(-g)}\ [\ (1 + A\varepsilon^2)R/2 - (\partial\varepsilon)^2/2 - (\lambda/4)(\varepsilon^2 - 1)^2\ ]}$$

Here ε is the Z-Spin scalar field, **A = 35/437** the geometric impedance, and $(\varepsilon^2-1)^2$ the Z_2-symmetry-breaking potential. The non-minimal coupling $(1+A\varepsilon^2)R$ bridges gravity and matter; the

kinetic term drives inflation; the potential provides the Z_2 attractor vacuum ($\varepsilon = \pm 1$). In its complex-scalar extension, the phase mode θ is exactly massless by Goldstone's theorem, and its gradient energy at galactic scales produces $\rho \propto 1/r^2$ isothermal haloes (the ε-Halo mechanism). Every prediction in the 55-paper series descends from this single action.

2.4. i-Tetration and the Bootstrap

The dynamics of the Z-sector are governed by imaginary-unit tetration—self-referential infinite rotation. The equation $z = i^z$ possesses a unique attracting fixed point:

$$\mathbf{z^* = (2i/\pi)\, W_0(-i\pi/2) = 0.43828 + 0.36059i} \quad \text{[PROVEN]}$$

The topological threshold $\eta_{topo} = |z^*|^2 = 0.3221$ structurally corresponds to the polyhedral face-counting matter fraction $\Omega_m = 38/121 = 0.3140$ (within 2.6%) [OBSERVATION]. Two independent mathematical routes—complex dynamics and combinatorial geometry—arrive at the same physical quantity: a self-consistent closure. The bootstrap programme (ZS-F0) connects the "how" ($z = i^z$) to the "why" through the chain: non-existence self-contradiction → Lawvere's fixed-point theorem → minimality of $\mathbb{C}$ → i-tetration.

§3. Key Predictions and Observational Verification: The Top-10 Scorecard

Observable	Z-Spin Prediction	Observed Value	Pull	Status
H_0 ratio	exp(A) = 1.0834	1.084 ± 0.018	0.05σ	**DERIVED**
Ω_m^eff	0.2908	0.2975 ± 0.0086	0.8σ	**DERIVED**
w_0	−1 (exact)	−1.055 ± 0.036	1.5σ	**DERIVED**
S_8	0.777	0.776 ± 0.017	0.06σ	**DERIVED**
Y_p	0.2410	0.2449 ± 0.004	−1.0σ	**DERIVED**
D/H (ΔN_eff incl.)	2.526×10^{-5}	$2.527\pm0.030\times10^{-5}$	−0.05σ	**DERIVED**
η_B	6.12×10^{-10}	$6.12\pm0.04\times10^{-10}$	0.02σ	**DERIVED**
n_s	0.9674	0.9649 ± 0.0042	0.6σ	**DERIVED**
r	0.0089	< 0.036 (BK18)	—	**DERIVED**
τ_D / τ_Penrose	12.49	Not yet measured	—	**TESTABLE**

Summary: All ten predictions with measured values lie within 2σ of observation. The baryon asymmetry η_B achieves the most precise agreement at 0.02σ. The decoherence-time ratio $\tau_D/\tau_{Penrose}$ is scheduled for decisive experimental verification by nanosphere interferometry in 2028–2032.

§4. Original Contributions: What Z-Spin Does for the First Time

4.1. Derivation of the Born Rule: From Postulate to Theorem

On the Q = 11 register, the block Laplacian's X–Y block vanishes identically ($L_{XY} \equiv 0$, PROVEN). The system (X) and the environment (Y) never communicate directly; every transition must traverse the Z-sector (dim = 2). Through Stinespring dilation, exactly dim(Z) = 2 Kraus operators are extracted and the CPTP condition is mathematically guaranteed, yielding the Born rule $p(x) = Tr(P_x \cdot \rho)$. Z-Spin is the *only* framework in the literature that derives, rather than postulates, the Born rule.

4.2. Resolution of the Hubble Tension: exp(A)

The geometric impedance A acts as a connection 1-form on the ε-field configuration space, and the Hubble ratio emerges as a Wilson loop on the polyhedral manifold: **H_0(local)/H_0(CMB) = exp(A) = 1.0834**. The predicted value 72.98 km/s/Mpc agrees with SH0ES (73.04 ± 1.04) to 0.06σ. The hierarchy is three-tiered: H_0(geo) = 64.81 → ×√(1+A) = 67.36 (Planck CMB) → ×exp(A) = 72.98 (local). A single constant, zero adjustment.

4.3. The Arrow of Time: Without the Past Hypothesis

The sector decomposition (Z, X, Y) = (2, 3, 6) combined with $L_{XY} \equiv 0$ generates a structural asymmetry: the Y-sector possesses twice the dimensionality of the X-sector, and every transition must pass through the Z-sector bottleneck (dim = 2). Each Z-mediated transition produces an entropy increment **ΔS = ln 2**. No new parameters, no past hypothesis. The arrow of time is a macroscopic manifestation of the (Z, X, Y) asymmetry.

4.4. Proton Lifetime and Cyclic Cosmology

The non-minimal coupling induces a topological phase drift δφ = A that accumulates over 78.45 cycles, triggering a topological transition. From the tunnelling action $S_{tunnel} = 5\pi/A = 196.13$, the proton lifetime is predicted as **$\tau_p \approx 2.56 \times 10^{34}$ yr** [DERIVED]—above the current Super-Kamiokande bound and within reach of Hyper-Kamiokande. The Z_2 vacuum transition ($\tau_6 = \exp(6\pi/A) \approx 2.78 \times 10^{51}$ yr) is 10^{17} times slower than proton decay, furnishing an infinite cyclic cosmology [PROVEN + DERIVED]. ZS-M12 v1.0 resolves the big-bang singularity via Auto-Surgery: the continuous i-tetration flow $d\Phi/d\tau = i^n - \Phi$ drives $\Phi \to z^*$ in ~3 τ_P, capping the conformal factor at $\Omega^2 = 1 + A \cdot \eta_topo = 1.0258$ [DERIVED]. A centrifugal barrier from Z-Telomere phase accumulation (ε_min ≈ 30.7 ≫ ε_sr = 2.64) naturally launches inflation [DERIVED-CONDITIONAL, contingent on V1; 35/35 PASS].

4.5. Gauge Coupling Constants and the Standard Model

The strong coupling constant $\alpha_s(M_Z) = 11/93 = 0.11828$ (PDG 2024: 0.1180 ± 0.0009, +0.31σ). The Weinberg angle $\sin^2\theta_W = (48/91) \times x^* = 0.23118$ (PDG: 0.23122 ± 0.00003, −1.26σ). The baryon-to-photon ratio $\eta_B = (6/11)^{35} = 6.117 \times 10^{-10}$ (Planck: $6.12 \pm 0.04 \times 10^{-10}$, +0.07σ). In the Higgs sector, the 1-loop quartic cancellation $STr(q^4) = 0$ is PROVEN, and the Higgs VEV is DERIVED via the Y-Sector Spectral VEV mechanism (v = 245.93 GeV, 0.12% from observation). All with zero free parameters.

4.6. Derivation of the Higgs VEV: From Geometry to the Electroweak Scale

In the Standard Model, the Higgs vacuum expectation value ($v = 246$ GeV) is a free parameter. Z-Spin derives it from geometry for the first time. The Y-Sector Spectral VEV (ZS-S4 v1.0):

$$\mathbf{v = M_P \times 2^{-418/9} \times 3^{-38/9} = 245.93\ GeV}$$

(observed: 246.22 GeV; deviation: 0.12%). The prime factors 2 and 3 in the exponents are the sole prime factors of $Q = 11$, and the Cheeger–Müller duality guarantees the spectral–topological correspondence. An anti-numerology Monte Carlo scan (500,000 random comparisons) yields $p = 0.028\%$. Z-Spin is the first framework to derive the Standard Model's most consequential free parameter from pure geometry [DERIVED].

4.7. Galactic ε-Haloes: Flat Rotation Curves Without Dark-Matter Particles

In the complex-scalar extension of the Z-Spin action, the phase mode θ is exactly massless by Goldstone's theorem. Its $\theta = 0$ solution yields a logarithmic profile $\theta(r) = \ln(r/r_s)/L$, whose gradient energy produces $\rho \propto 1/r^2$ isothermal haloes—reproducing flat rotation curves without invoking dark-matter particles [DERIVED]. The supermassive black hole at the galactic centre serves as a Z-anchor ($|\Phi|(0) = 0$, $\pi_1(U(1)) = \mathbb{Z}$), the source of spatial coherence, while the halo boundary r_Z marks the decoherence frontier where the θ-gradient vanishes. Gaia DR3 confirms $r_Z(MW) \approx 18.3$ kpc (observed ~19 kpc, 4% agreement). A single action yields eight galactic observables: flat rotation curves, the BTFR (slope 4), MOND $a_0 = cH_0/6$, the M–σ relation ($\beta = 4$), and elliptical-galaxy vortex glasses. If a dark-matter particle is detected, the ε-Halo mechanism is immediately falsified (Gate F-A5.7).

4.8. QCD Confinement and the Glueball Mass: The Spinor Mass Gap

ZS-S7 derives the QCD confinement scale Λ_QCD and the scalar glueball mass m(0++) from the Z-Spin action with zero free parameters. The Hodge 2-form Laplacian L_2 on the Y-sector truncated icosahedron ($V = 60$, $F = 32$) has spectral gap $\lambda_1 = 1.2428$. The Spinor-Descartes-Euler identity unifies three independent results: Descartes' total deficit angle (4π), Euler's topological invariant ($2\pi\chi$), and the SU(2) spinor period (4π), yielding $4\pi = 2\pi \cdot \dim(Z)$ = spinor period [PROVEN]. Combined with vertex-transitivity, this gives a uniform per-vertex vacuum source $E_local = vA/V_Y = 328$ MeV. The spectral equilibrium condition yields $\Lambda_QCD = vA/(\lambda_1 V_Y) = 264$ MeV (lattice: 260 ± 20 MeV, +0.2σ) [DERIVED-CONDITIONAL].

A remarkable topological cancellation then eliminates all Y-sector internal structure from the glueball mass formula: $m(0{+}{+}) = vA/Q = 245.93 \times (35/437) / 11 = 1.791$ GeV (lattice: 1.73 ± 0.05 GeV, +1.2σ) [DERIVED-CONDITIONAL]. This cancellation is the Hodge-theoretic analog of the Gauss-Bonnet theorem: local dynamics cancel, leaving only topological invariants (v, A, Q). Additional exact results include $b_0 = (V+F)_Y/G = 23/3$ (SM β-function coefficient at $n_f = 5$, exact match, PROVEN) and $n_f = V_Y/G = 60/12 = 5$ (exact, DERIVED). The mass gap is not a dynamical accident but a geometric necessity rooted in the spinor nature of the Z-sector mediator.

§5. Falsification Conditions and Outlook

Z-Spin supplies concrete, pre-registered gates by which it can be definitively refuted:

Falsification Gate	Predicted Value	Timeline	Experiment
n_s, r	$n_s = 0.9674$, $r = 0.0089$ (distinguishable from Starobinsky at 6σ)	2031–34	CMB-S4 / LiteBIRD
$\tau_D / \tau_{Penrose}$	$12.49 \pm 20\%$ (decisive test)	2028–32	Nanosphere interferometry
τ_p	2.56×10^{34} yr	2027–	Hyper-Kamiokande
δ_{CP}	$\pm\pi/2 + \arctan(A) = \pm 94.6°$	2028–	DUNE / T2HK
w_0	-1 (exact); reject at 3σ deviation	2025–	DESI / Euclid
ΔN_{eff}	$2A = 0.160$	2028–30	CMB-S4
DM particle	Detection $\Rightarrow$ ε-Halo falsified	Ongoing	Direct-detection experiments

§6. Closing Remarks

Z-Spin Cosmology interprets the six crises of modern physics as manifestations of a shared geometric origin. Starting from a single geometric constant **A = 35/437**, governed by a single action, the framework predicts observables spanning sixty orders of magnitude, derives the Born rule, and accounts for the arrow of time—all with zero free parameters.

Whether the fundamental constants of nature are products of geometric necessity or an unexplained collection of coincidences—this is a question that only nature can settle. Z-Spin Cosmology poses that question in the language nature respects: concrete numbers and falsifiable conditions. With rigour and humility in equal measure, we await nature's verdict.

— Z-Spin Cosmology Collaboration, 2026

Acknowledgements & Code Availability

Acknowledgements. This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/mpmath, 50-digit precision) is publicly available.

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v7.0)v1.0 update (March 2026): Synchronized with The Book v1.0. Key changes: (1) Paper count: 46 → 52 (ZS-F6, ZS-M8–M12 integrated). (2) Face counting: Ω_m = 38/121 (was 39/121), Ω_m^eff = 0.2908, S_8 = 0.777. (3) §2.2: ZS-F6 self-consistency proof added. (4) §4.4: ZS-M12 Auto-Surgery and centrifugal barrier added. (5) Scorecard updated for face counting. No prior content deleted.v1.0 update (April 2026): Synchronized with The Book v1.0 (55-paper release). Key changes: (1) Paper count: 52 → 55 (ZS-F7, ZS-M13, ZS-S7 integrated). (2) NEW §4.8: QCD Confinement and the Glueball Mass (ZS-S7, 18/18 PASS): Λ_QCD = 264 MeV (+0.2σ), m(0++) = vA/Q = 1.791 GeV (+1.2σ), topological cancellation theorem, b_0 = 23/3, n_f = 5 exact. (3) ZS-F7: Reuleaux Spectral Geometry (18/18 PASS). ZS-M13: Arithmetic Spectral Correspondence (30/30 PASS). (4) Verification: ~155 gates, ~1,466+ tests. No prior content deleted.

Z-Spin GitHub

https://github.com/KennyKang-git/zspin